



VP AIR TEST NEET (2025-26)

PART TEST - 04

DROPPER

DURATION: 180 Minutes

Date: 12-10-2025

M. MARKS: 720

Topics Covered

Physics:	Rotational Motion, Gravitation and Mechanical Properties of Solids
Chemistry:	Thermodynamics and Chemical Equilibrium
Botany:	Molecular Basis of Inheritance, The Living world
Zoology:	Excretory Products and their Elimination

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

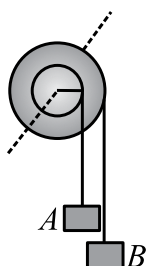
Name of the Student (In CAPITALS): _____

Roll Number: _____

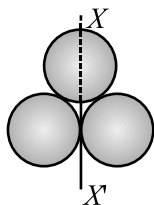
OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Figure shows a small wheel fixed coaxially on a bigger one of double the radius of smaller wheel. The system rotates uniformly about the common axis. The strings supporting A and B do not slip on the wheels. If x and y be the distances travelled by A and B in the same time interval, then:



- (1) $x = 2y$ (2) $x = y$
 (3) $y = 2x$ (4) $2x = 3y$
2. Wires A and B are made from the same material. A has twice the diameter and three times the length to that of B . If the elastic limits are not reached, when each wire is stretched by the same tension, ratio of energy stored in wire A and B is:
- (1) 2 : 3 (2) 3 : 4
 (3) 3 : 2 (4) 6 : 1
3. Assuming the earth to be a sphere of uniform mass density, the weight of a body at a depth $d = \frac{R}{2}$ from the surface of earth, if its weight on the surface of earth is 200 N, will be: (Given R = Radius of earth)
- (1) 400 N (2) 500 N
 (3) 300 N (4) 100 N
4. Three solid spheres of mass M and radius R are shown in the figure. The moment of inertia of the system about XX' axis will be:



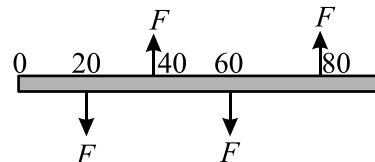
- (1) $\frac{7}{2}MR^2$ (2) $\frac{14}{5}MR^2$
 (3) $\frac{16}{5}MR^2$ (4) $\frac{21}{5}MR^2$
5. Consider a planet of same size of earth but its mass is 0.2 times the mass of earth. If R_e is the maximum range of a projectile on the earth's surface, what is the maximum range on the surface of the planet for the same velocity of projection:
- (1) $0.2 R_e$ (2) $2 R_e$
 (3) $0.5 R_e$ (4) $5 R_e$

6. Three particle each of mass m are placed at the corners of an equilateral triangle of side d . If the side of the triangle is changed from d to $2d$, then work done by external agent on this system is:

(1) $-\frac{3Gm^2}{2d}$ (2) $\frac{3Gm^2}{2d}$
 (3) $\frac{Gm^2}{2d}$ (4) $-\frac{3Gm^2}{d}$

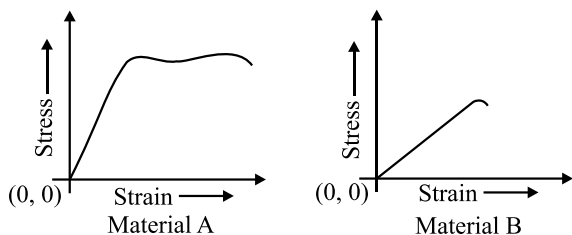
7. Two satellites A and B of the same mass are orbiting the earth at altitudes R and $3R$ respectively (R is radius of the earth). Find the ratio of their respectively kinetic energy is:
- (1) 1 : 1 (2) 2 : 1
 (3) 4 : 1 (4) 8 : 1

8. Four equal and parallel forces are acting on a rod (as shown in figure) at distances of 20 cm, 40 cm, 60 cm and 80 cm respectively from one end of the rod. Under the influence of these forces the rod:



- (1) is at rotational as well as translational equilibrium.
 (2) experiences a torque
 (3) experiences a linear acceleration
 (4) experiences a torque and also a linear acceleration
9. Four identical hollow cylindrical columns of mild steel support a big structure of mass 50×10^3 kg. The inner and outer radii of each column are 50 cm and 100 cm respectively. Assuming uniform load distribution, calculate the compression strain of each column. [Use $Y = 2.0 \times 10^{11}$ Pa, $g = 10$ m/s²]
- (1) 3.60×10^{-8} (2) 2.65×10^{-7}
 (3) 1.87×10^{-3} (4) 7.07×10^{-4}
10. A wheel starts rotating from rest and attains an angular velocity of 60 rad/sec in 5 second under uniform angular acceleration. The total angular displacement in radians will be:
- (1) 60 (2) 80
 (3) 100 (4) 150
11. A string is wrapped around the rim of a wheel of moment of inertia 0.40 kgm² and radius 10 cm. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a force of 40 N. The angular velocity of the wheel after 10 s is x rad/s, where x is:
- (1) 80 (2) 100
 (3) 120 (4) 150

12. The stress-strain graph for two material A and B are shown in figure. The graph are drawn to the same scale. Select the **correct** statement:



Statement-A: Material B has greater young's modulus than A

Statement-B: Material A is more ductile than material B

Statement-C: Material A is less ductile than material B

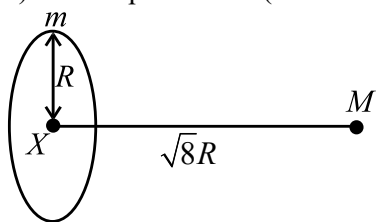
Statement-D: Material A has greater Young's modulus than B

- (1) C and D only (2) B and D only
(3) A and B only (4) A and C only

13. Four particles each of mass m are placed at the corners of a square of side length ℓ . The radius of gyration of the system about an axis perpendicular to the square and passing through centre is:

- (1) $\frac{\ell}{\sqrt{2}}$ (2) $\frac{\ell}{2}$
(3) ℓ (4) $\ell\sqrt{2}$

14. Find the gravitational force of attraction between the uniform ring and a point mass as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centers $\sqrt{8}R$ is the distance between the centers of the ring (of mass m) and the point mass (of mass M).



- (1) $\frac{\sqrt{8}}{9} \cdot \frac{GmM}{R}$ (2) $\frac{2\sqrt{2}}{3} \cdot \frac{GmM}{R^2}$
(3) $\frac{1}{3\sqrt{8}} \cdot \frac{GmM}{R^2}$ (4) $\frac{\sqrt{8}}{27} \cdot \frac{GmM}{R^2}$

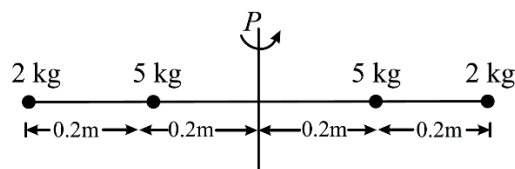
15. A satellite of mass m is circulating around the earth with constant angular velocity. If radius of the orbit is R_0 and mass of the earth M , the angular momentum about the centre of the earth is:

- (1) $m\sqrt{GMR_0}$ (2) $M\sqrt{GmR_0}$
(3) $m\sqrt{\frac{GM}{R_0}}$ (4) $M\sqrt{\frac{Gm}{R_0}}$

16. A body starts from rest at a point at distance R_0 , from the centre of the earth of mass M and radius R . The velocity acquired by the body when it reaches the surface of the earth will be:

- (1) $GM\left(\frac{1}{R} - \frac{1}{R_0}\right)$
(2) $2GM\left(\frac{1}{R} - \frac{1}{R_0}\right)$
(3) $\sqrt{2GM\left(\frac{1}{R} - \frac{1}{R_0}\right)}$
(4) $2GM\sqrt{\left(\frac{1}{R} - \frac{1}{R_0}\right)}$

17. Four masses are fixed on a massless rod as shown in figure. The moment of inertia about the axis P is:



- (1) 2 kg m^2 (2) 1.04 kg m^2
(3) 1.52 kg m^2 (4) 0.3 kg m^2

18. On the x -axis and at a distance x from the origin, the gravitational field due to a mass distribution is given by $\frac{2A}{x^3}$ in the x -direction. The magnitude of gravitational potential on the x -axis at a distance x , taking its value to be zero at infinity, is: (A is constants)

- (1) Ax (2) $\frac{A}{x^2}$
(3) $\frac{A}{x^3}$ (4) Ax^2

19. A satellite of 10^3 kg mass is revolving in circular orbit of radius $2R$. If $\frac{10^4 R}{6} \text{ J}$ energy is supplied to the satellite, it would revolve in a new circular orbit of radius: (use $g = 10 \text{ m/s}^2$, $R = \text{radius of earth}$)

- (1) $2.5 R$ (2) $3 R$
(3) $4 R$ (4) $6 R$

20. The maximum and minimum distances of a comet from the Sun are $1.6 \times 10^{12} \text{ m}$ and $8.0 \times 10^{10} \text{ m}$ respectively. If the speed of the comet at the nearest point is $6 \times 10^4 \text{ m s}^{-1}$, the speed at the farthest point is:

- (1) $3.0 \times 10^3 \text{ m/s}$ (2) $1.5 \times 10^3 \text{ m/s}$
(3) $4.5 \times 10^3 \text{ m/s}$ (4) $6.0 \times 10^3 \text{ m/s}$

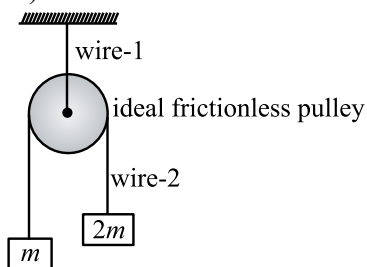
21. A block of mass 500 kg is suspended by wire of length 70 cm. The area of cross-section of wire is 10 mm^2 . When the load is removed, the wire contracts by 0.5 cm. The Young's modulus of the material of wire will be: ($g = 10 \text{ m/s}^2$)

(1) $10 \times 10^{14} \text{ N/m}^2$ (2) $4 \times 10^{14} \text{ N/m}^2$
 (3) $8 \times 10^{11} \text{ N/m}^2$ (4) $7 \times 10^{10} \text{ N/m}^2$

22. Two stars of masses m and $2m$ at a distance d rotate about their common center of mass in free space. The period of revolution is:

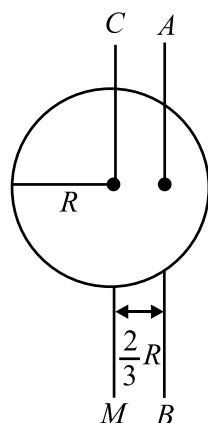
(1) $\frac{1}{2\pi} \sqrt{\frac{d^3}{3Gm}}$ (2) $2\pi \sqrt{\frac{d^3}{3Gm}}$
 (3) $\frac{1}{2\pi} \sqrt{\frac{3Gm}{d^3}}$ (4) $2\pi \sqrt{\frac{3Gm}{d^3}}$

23. Find the ratio of stress present in wire-1 to that in wire-2 shown in the diagram while both wires have same cross section area. (assume both wires are massless)



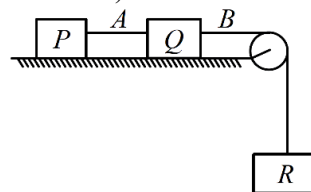
(1) 1 (2) 2
 (3) 3 (4) 4

24. I_{CM} is the moment of inertia of a circular disc about an axis (CM) passing through its center and perpendicular to the plane of the disc. I_{AB} is its moment of inertia about an axis AB perpendicular to plane and parallel to the axis CM at a distance $\frac{2}{3}R$ from the center (where R is the radius of the disc). The ratio of I_{AB} and I_{CM} is $x : 9$. The value of x is:



(1) 17 (2) 22
 (3) 35 (4) 48

25. Each of three blocks P , Q and R shown in figure have a mass of 3 kg. Each of the wire A and B has cross-sectional area 0.005 cm^2 and Young's modulus $2 \times 10^{11} \text{ N m}^{-2}$. Neglecting friction, the longitudinal strain in wire B is $\text{_____} \times 10^{-4}$. (Take $g = 10 \text{ m/s}^2$)



(1) 2 (2) 5
 (3) 9 (4) 12

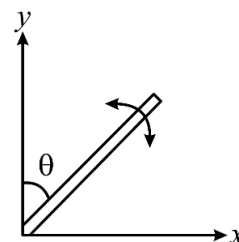
26. Two persons pull a wire towards themselves, each holding it from opposite ends. Each person exerts a force of 200 N on the wire. Young's modulus of the material of wire is $1 \times 10^{11} \text{ N m}^{-2}$. Original length of the wire is 2 m and the area of cross-section is 2 cm^2 . The wire will extend in length by $\text{_____} \mu\text{m}$.

(1) 15 (2) 20
 (3) 25 (4) 30

27. The moment of inertia and rotational kinetic energy of a fly wheel are 20 kg-m^2 and 1000 joule respectively. Its angular frequency (in rev/min) would be:

(1) $\frac{600}{\pi}$ (2) $\frac{25}{\pi^2}$
 (3) $\frac{5}{\pi}$ (4) $\frac{300}{\pi}$

28. A slender uniform rod of mass M and length ℓ is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is:

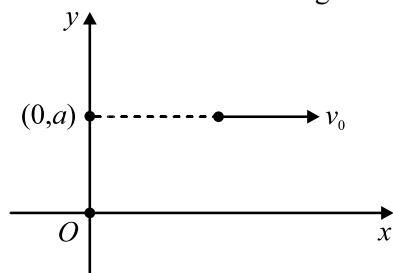


(1) $\frac{3g}{2l} \cos \theta$ (2) $\frac{2g}{3l} \cos \theta$
 (3) $\frac{3g}{2l} \sin \theta$ (4) $\frac{2g}{3l} \sin \theta$

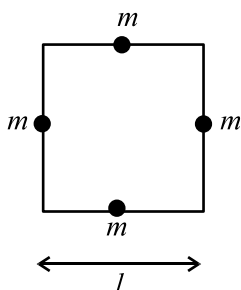
29. The Young's modulus of a wire of length L and radius r is Y . If the length and radius are reduced to $\frac{L}{2}$ and $\frac{r}{2}$ respectively, then its Young's modulus will become:

(1) $\frac{Y}{2}$ (2) Y
 (3) $2Y$ (4) $4Y$

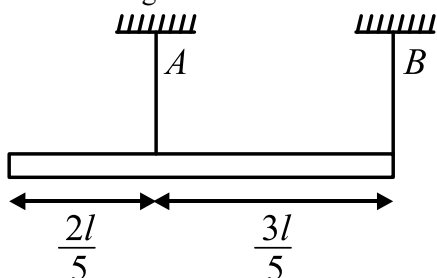
30. A point mass m is moving with velocity v_0 along the x -direction as shown in figure. Find the angular momentum of mass m about origin O .



- (1) $mv_0a(\hat{k})$ (2) $-\frac{mv_0a}{3}\hat{k}$
 (3) $-\frac{mv_0a}{2}\hat{k}$ (4) $mv_0a(-\hat{k})$
31. Four particles each of mass m are kept fixed at the center of sides of a square of edge length l as shown below. Find the magnitude of gravitational force on any one mass due to the remaining masses.

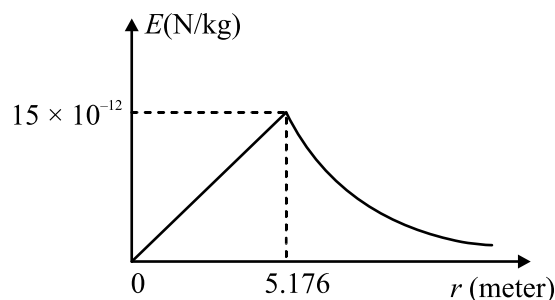


- (1) $\frac{Gm^2}{l^2}(\sqrt{2}+1)$ (2) $\frac{Gm^2}{l^2}(2\sqrt{2}+1)$
 (3) $\frac{Gm^2}{l^2}\left(1+\frac{3}{\sqrt{2}}\right)$ (4) $\frac{Gm^2}{l^2}\left(1+\frac{1}{\sqrt{2}}\right)$
32. A uniform rod of mass M and length L is hung vertically from the ceiling. What is the total elongation in the rod due to its own weight? (Young's modulus is Y , area of cross-section is A and g is acceleration due to gravity)
- (1) Zero (2) $\frac{MgL}{AY}$
 (3) $\frac{MgL}{2AY}$ (4) $\frac{2MgL}{AY}$
33. The ratio of tensions in string A and B supporting a uniform long horizontal rod of mass M and length l as shown in figure is:

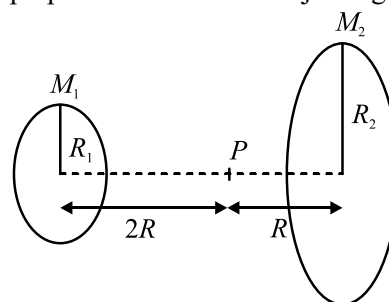


- (1) 2 : 3 (2) 2 : 5
 (3) 3 : 2 (4) 5 : 1

34. Variation of magnitude of gravitational field (E) with distance (r) from the centre of uniform solid sphere is shown in figure. The mass of solid sphere is: ($G = 6.7 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $(5.176)^2 = 26.8$)

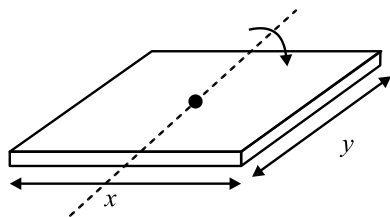


- (1) 2 kg (2) 3 kg
 (3) 5 kg (4) 6 kg
35. Consider a Disc of mass 5 kg, radius 2 m, rotating with angular velocity of 10 rad/s about an axis perpendicular to the plane of rotation. An identical disc is kept gently over the rotating disc along the same axis. The energy dissipated so that both the discs continue to rotate together without slipping is _____ J.
- (1) 180 (2) 210
 (3) 250 (4) 290
36. Position vector $\vec{r}$ of a particle of mass $m = 1 \text{ kg}$ is given by $\vec{r} = (t\hat{i} + 3t^2\hat{j}) \text{ m}$ (where t is in second). Its angular momentum (in $\text{kg m}^2/\text{s}$) with respect to origin at $t = 1 \text{ s}$ is:
- (1) $3\hat{k}$ (2) $-3\hat{k}$
 (3) 0 (4) $6\hat{k}$
37. Two rings of radius $R_1 = R$ and $R_2 = 2R$ with common central axis is shown in figure. If net gravitational field at point P , which is on the line joining between their centres is zero, then ratio of their masses $\left(\frac{M_1}{M_2}\right)$ is: (Plane of ring is perpendicular to the line joining their centres)

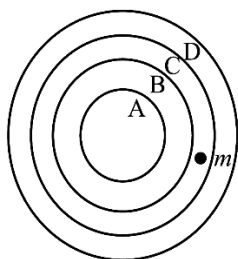


- (1) $\frac{1}{2}$ (2) 1
 (3) 3 (4) $\frac{1}{4}$

38. A thin uniform rectangular plate is rotating about an axis which is in the plane of plate, passing through its centre and perpendicular to side x as shown in figure. The radius of gyration of the plate about this axis is:



- (1) $2\sqrt{3}x$ (2) $\frac{x}{2\sqrt{3}}$
 (3) $2\sqrt{3}y$ (4) $\frac{y}{2\sqrt{3}}$
39. A rigid body can be hinged about any point on the x -axis. When it is hinged such that the hinge is at x , the moment of inertia is given by $I = 2x^2 - 12x + 27$. The x -coordinate of centre of mass is:
- (1) $x = 2$ (2) $x = 0$
 (3) $x = 1$ (4) $x = 3$
40. Four concentric uniform spherical shells A, B, C, D of different radius are shown in the figure. A particle of mass m is placed somewhere in the region between the shells B and C . The gravitational force due to which of the shells on the mass m is non-zero.



- (1) C and D only
 (2) only D only
 (3) A and B only
 (4) All the shells A, B, C and D
41. Select the **correct** statement from the following.
- (1) The orbital velocity of a satellite increases with the radius of the orbit
 (2) Escape velocity of a particle from the surface of the earth depends on the mass of the particle at which it is fired
 (3) The time period of a satellite does not depend on the radius of the orbit
 (4) The orbital velocity is inversely proportional to the square root of the radius of the orbit

42. If R_E be the radius of Earth, then the ratio between the acceleration due to gravity at a depth ' r ' below and a height ' r ' above the earth surface is:

(1) $1 - \frac{r}{R_E} - \frac{r^2}{R_E^2} - \frac{r^3}{R_E^3}$
 (2) $1 + \frac{r}{R_E} + \frac{r^2}{R_E^2} + \frac{r^3}{R_E^3}$
 (3) $\left(1 - \frac{2r}{R_E}\right) \left(1 + \frac{2r}{R_E}\right)^2$
 (4) $\left(1 - \frac{r}{R_E}\right) \left(1 + \frac{r}{R_E}\right)^2$

43. A uniform bar of length ' L ' and cross-sectional area ' A ' is subjected to a tensile load ' F '. ' Y ' be the Young's modulus and ' σ ' be the Poisson's ratio then volumetric strain is:

(1) $\frac{F}{AY}(1 - \sigma)$
 (2) $\frac{F}{AY}(2 - \sigma)$
 (3) $\frac{F}{AY}(1 - 2\sigma)$
 (4) $\frac{F}{AY} \cdot \sigma$

44. A particle is at a distance r from the axis of rotation. A given torque τ produces some angular acceleration in it. If the mass of the particle is doubled and its distance from the axis is halved, the value of torque to produce the same angular acceleration is:

(1) $\frac{\tau}{2}$ (2) τ
 (3) 2τ (4) 4τ

45. Match **List-I** (Physical Quantities) with **List-II** (SI Units).

List-I		List-II	
(A)	Angular acceleration	(I)	Newton-m
(B)	Torque	(II)	rad/sec ²
(C)	Angular momentum	(III)	kg.m ²
(D)	Moment of Inertia	(IV)	kg.m ² /sec

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
 (2) A-III, B-I, C-IV, D-II
 (3) A-II, B-III, C-IV, D-I
 (4) A-I, B-II, C-IV, D-III

46. Given below are two statements:

Statement I: For every chemical reaction at equilibrium, standard Gibbs energy of reaction is zero.

Statement II: The endothermic reactions are favoured at lower temperature and the exothermic reactions are favoured at higher temperature.

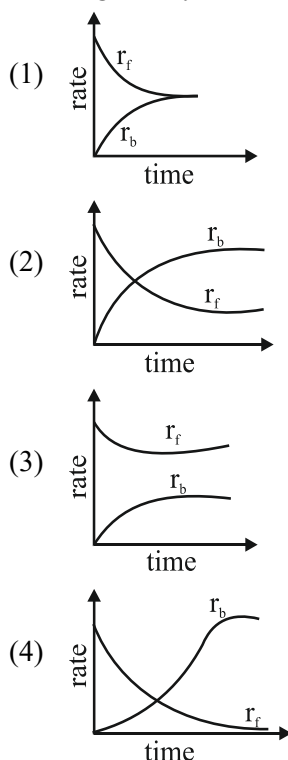
In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

47. The molar heat capacity of a gas at constant volume is $12 \text{ J K}^{-1} \text{ mol}^{-1}$. The change in internal energy is 4000 J upon heating it from 300 K to 500 K. The number of moles of the gas is:

- (1) 1.67
- (2) 1.24
- (3) 2.02
- (4) 0.52

48. For the reversible reaction $A \rightleftharpoons B$, the variation of rate of forward (r_f) and backward (r_b) reactions with time is given by:



49. $(\Delta H - \Delta U)$ for the formation of one mole of CO(g) from its elements at 298 K is:
(Given: $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

- (1) 0.67 kJ
- (2) 1.24 kJ
- (3) 2.1 kJ
- (4) 1.81 kJ

50. At a certain temperature, only 50% of AB dissociates as $2\text{AB(g)} \rightleftharpoons \text{A}_2\text{(g)} + \text{B}_2\text{(g)}$. The equilibrium constant of the reaction is:

- (1) 0.50
- (2) 0.25
- (3) 0.75
- (4) 1.25

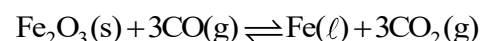
51. Match List-I with List-II.

List-I (Thermodynamic quantity)		List-II (Partial derivatives)	
(A)	C_p	(I)	$\left(\frac{dU}{dT}\right)_V$
(B)	S	(II)	$\left(\frac{dH}{dT}\right)_P$
(C)	C_v	(III)	$-\left(\frac{dG}{dT}\right)_P$
(D)	V	(IV)	$\left(\frac{dG}{dP}\right)_T$

Choose the **correct** answer from the options given below:

- (1) A-IV; B-I; C-II; D-III
- (2) A-II; B-I; C-IV; D-III
- (3) A-II; B-III; C-I; D-IV
- (4) A-IV; B-II; C-I; D-III

52. Which of the following changes will **not** disturb the given equilibrium?



- A. Addition of $\text{CO}_2(\text{g})$
- B. Removal of $\text{Fe}(\text{l})$
- C. Removal of $\text{CO}(\text{g})$
- D. Addition of $\text{Fe}_2\text{O}_3(\text{s})$

- (1) A and C
- (2) C and D
- (3) B and D
- (4) B and A

53. Given below are two statements:

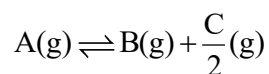
Statement I: First law of thermodynamics is not adequate in predicting the direction of a process.

Statement II: The heat absorbed during the isothermal expansion of an ideal gas against vacuum is zero.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

54. Consider the equilibrium,



the **correct** relationship between K_p , α and equilibrium pressure P is:

(Where, degree of dissociation (α) $\ll 1$)

(1) $K_p = \frac{P^{1/2} \alpha^{1/2}}{2^{3/2}}$

(2) $K_p = \frac{P^{1/2} \alpha^{3/2}}{2^{3/2}}$

(3) $K_p = \frac{P^{1/2} \alpha^{3/2}}{2^{1/2}}$

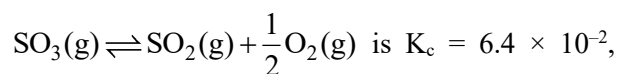
(4) $K_p = \frac{P^{1/2} \alpha^{1/2}}{2^{1/2}}$

55. How many among the following are intensive quantities?

Enthalpy, Temperature, Volume, Refractive index, Specific heat capacity, Entropy, Molarity

- (1) 6
- (2) 4
- (3) 5
- (4) 3

56. If the equilibrium constant for the reaction



then the equilibrium constant for the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, will be:

- (1) 416.84
- (2) 311.63
- (3) 244.14
- (4) 368.11

57. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Temperature is the measure of average chaotic motion of particles in the system.

Reason R: A system at higher temperature has greater randomness in it than one at lower temperature.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

58. Given below are two statements:

Statement I: When a system containing ice in equilibrium with water (liquid) is heated, heat is absorbed by the system and there is no change in the temperature of the system until whole ice gets melted.

Statement II: A catalyst alters the equilibrium constant (K_c) of the reaction.

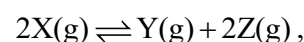
In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

59. The ΔH and ΔS for the reaction $2Cl(g) \rightarrow Cl_2(g)$, respectively are:

- (1) < 0 and < 0
- (2) > 0 and < 0
- (3) > 0 and > 0
- (4) < 0 and > 0

60. 8 mol of $X(g)$ are introduced into a 1 L flask which dissociates as:



at equilibrium, 2 mol of $Y(g)$ are present. The equilibrium constant of the reaction is:

- (1) 4
- (2) 8
- (3) 2
- (4) 6

61. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Energy can neither be created nor be destroyed.

Reason R: The energy of an isolated system is constant.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

62. In which of the following, a reaction will proceed almost towards completion?

- (1) $K_{eq} = 10^5$
- (2) $K_{eq} = 10^{-2}$
- (3) $K_{eq} = 10^{-5}$
- (4) $K_{eq} = 1$

63. The species which by definition has zero standard molar enthalpy of formation at 298 K is:

- (1) Cl(g)
- (2) S(rhombic)
- (3) Br₂(g)
- (4) C(diamond)

64. An example of reversible reaction is:

- (1) $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$
- (2) $2\text{K}(\text{s}) + \text{H}_2\text{O}(\ell) \rightarrow 2\text{KOH}(\text{aq}) + \text{H}_2(\text{g})$
- (3) $\text{Pb}(\text{NO}_3)_2(\text{aq}) + 2\text{NaI}(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{NaNO}_3(\text{aq})$
- (4) $\text{KNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{KCl}(\text{aq}) + \text{NaNO}_3(\text{aq})$

65. Enthalpy of which reaction represents enthalpy of atomization?

- (1) $\text{N}_2\text{H}_4(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2(\text{g})$
- (2) $\frac{1}{2}\text{H}_2(\text{g}) \rightarrow \text{H}(\text{g})$
- (3) $\text{C}(\text{s, diamond}) \rightarrow \text{C}(\text{s, graphite})$
- (4) $\text{CH}_4(\text{g}) \rightarrow \text{C}(\text{g}) + 4\text{H}(\text{g})$

66. For the reaction,



At constant temperature, the forward reaction is favoured by:

- (A) adding inert gas at constant pressure
- (B) adding PCl₅ at constant volume
- (C) increasing the volume of container

Choose the *most appropriate* option:

- (1) A and B only
- (2) B and C only
- (3) A and C only
- (4) A, B and C

67. Consider the following statements:

- (A) A spontaneous process is an irreversible process.
- (B) Heat has randomising influence on the system.
- (C) It is impossible to determine lattice enthalpies directly by experiment.
- (D) For chemical reactions, heat absorbed at constant pressure, is measured in a bomb calorimeter.
- (E) When a system is in equilibrium, the entropy is maximum.

The **correct** statements are:

- (1) A, B, C and E only
- (2) A, B and E only
- (3) A, B and D only
- (4) B, C and D only.

68. **Correct** statement(s) among the following is/are:

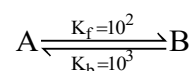
- A. All Chemical equilibria are static in nature.
- B. Molar Concentration of pure solid or liquid is constant
- C. A Catalyst lowers the activation energy for the forward and reverse reactions by exactly the same amount.

- (1) A and B only
- (2) B and C only.
- (3) A and C only
- (4) A, B and C

69. The enthalpy change for a reaction does **not** depend upon:

- (1) the physical states of reactants.
- (2) use of different reactants for same products.
- (3) the differences in initial and final temperatures of reacting substances.
- (4) the number of intermediate reaction steps.

70. For the reaction



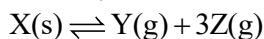
equilibrium constant is:

- (1) 1
- (2) 0.1
- (3) 10
- (4) 100

71. Which of the reaction defines $\Delta_f H^\circ$?

- (1) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
- (2) $C(g) + 2H_2(g) \rightarrow CH_4(g)$
- (3) $\frac{3}{2}O_2(g) \rightarrow O_3(g)$
- (4) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

72. If the concentration of Z at equilibrium in the reaction;



is decreased by $\left(\frac{1}{2}\right)$ times, then the equilibrium

concentration of Y(g):

- (1) increases by 8 times
- (2) increases by 64 times
- (3) decreases by 4 times
- (4) decreases by 64 times

73. A process will be non-spontaneous at all temperatures when:

- (1) $\Delta H < 0$ and $\Delta S < 0$
- (2) $\Delta H > 0$ and $\Delta S > 0$
- (3) $\Delta H > 0$ and $\Delta S < 0$
- (4) $\Delta H < 0$ and $\Delta S > 0$

74. Match List-I with List-II.

List-I		List-II	
(A)	If $Q_c > K_c$	(I)	Reaction proceeds rarely
(B)	If $Q_c < K_c$	(II)	Reaction proceeds nearly to completion
(C)	If $K_c < 10^{-3}$	(III)	Reaction will proceed in the direction of products
(D)	If $K_c > 10^3$	(IV)	Reaction will proceed in the direction of reactants

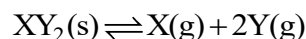
Choose the **correct** answer from the options given below:

- (1) A-III; B-IV; C-I; D-II
- (2) A-IV; B-III; C-I; D-II
- (3) A-IV; B-III; C-II; D-I
- (4) A-III; B-IV; C-II; D-I

75. The enthalpy of vaporisation of a liquid is 40 kJ mol^{-1} and the entropy of vaporization is $60 \text{ J mol}^{-1} \text{ K}^{-1}$. The boiling point of liquid at 1 atm is:

- (1) 333.34 K
- (2) 666.67 K
- (3) 546.43 K
- (4) 433.12 K

76. Consider the decomposition of XY_2 :



If the equilibrium pressure is 12 atm, the value of K_p for the reaction is:

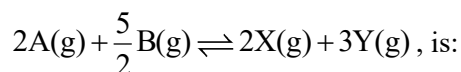
- (1) 64 atm³
- (2) 256 atm³
- (3) 128 atm³
- (4) 96 atm³

77. If the value of $\left(\frac{C_p}{C_v}\right)$ for a gas is 1.4, then the gas

may be:

- (1) He
- (2) O₂
- (3) SO₂
- (4) NO₂

78. Unit of K_c for the equilibrium



- (1) $\text{mol}^{\frac{1}{2}} \text{L}^{\frac{1}{2}}$
- (2) mol L^{-1}
- (3) $\text{mol}^{-\frac{3}{2}} \text{L}^{\frac{3}{2}}$
- (4) $\text{mol}^{-2} \text{L}^2$

79. Consider the following processes:

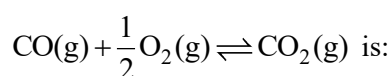
Process ΔH (kJ mol⁻¹)

- $\frac{1}{2}A \rightarrow B$ +100 ... (1)
- $3B \rightarrow 2C + D$ - 120 ... (2)
- $E + A \rightarrow 2D$ +300 ... (3)

For, $B + D \rightarrow E + 2C$, ΔH will be:

- (1) -180 kJ mol⁻¹
- (2) +160 kJ mol⁻¹
- (3) -220 kJ mol⁻¹
- (4) +240 kJ mol⁻¹

80. The ratio $\frac{K_c}{K_p}$ for the reaction



- (1) RT
- (2) $\frac{1}{\sqrt{RT}}$
- (3) $\sqrt{RT}$
- (4) 1

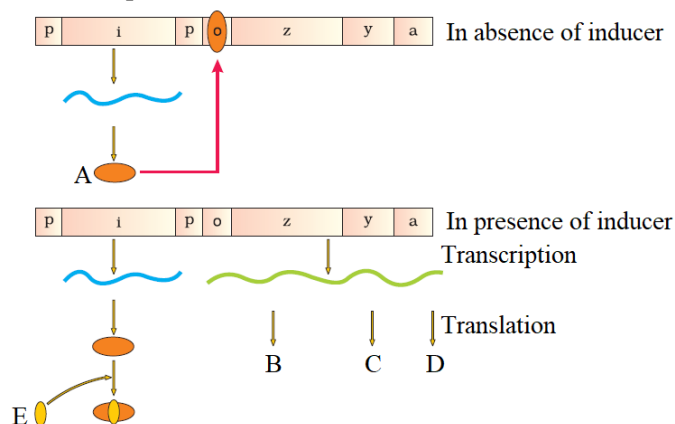
81. The enthalpy change for the given reaction (in kJ mol^{-1}) will be:
 $\text{CH}_2=\text{CH}_2 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$
 (Given: Bond energy of C–C, C=C, C–H and H–H are x, y, z and w kJ mol^{-1} respectively):
 (1) $y + w - 2z - x$
 (2) $2z - y + w + x$
 (3) $y + x - z - 2w$
 (4) $z + 2x - y - w$
82. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: When an inert gas is added to the equilibrium, $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ at constant pressure, the equilibrium will not disturb.
Reason R: In the reaction, $\Delta n_g = 0$.
 In the light of above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true and R is NOT the correct explanation of A.
83. The internal energy change when one mole of water is vapourised at 1 bar pressure at 100°C will be:
 (Given: $\Delta_{\text{vap}}H$ for water = 41 kJ mol^{-1})
 (1) 43.5 kJ mol^{-1}
 (2) 28.1 kJ mol^{-1}
 (3) 37.9 kJ mol^{-1}
 (4) 52.6 kJ mol^{-1}
84. Given below are two statements:
Statement I: At melting point of ice, there is absorption of heat in order to overcome intermolecular forces of attraction within the molecules of water in ice and kinetic energy of molecules is increased at melting point.
Statement II: A catalyst can alter the equilibrium constant (K_c) of the reaction at constant temperature.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.
85. Two moles of an ideal gas expands isothermally and reversibly from 5 L to 10 L at 300 K. The work done in the process respectively are:
 (1) -2.87 kJ
 (2) -4.12 kJ
 (3) -2.16 kJ
 (4) -3.46 kJ
86. For the equilibrium $2\text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$, the value of ΔG° at 298K is:
 (1) $2.303 \times 14 \times R \times 298 \text{ J mol}^{-1}$
 (2) $14 \times R \times 298 \text{ J mol}^{-1}$
 (3) $7 \times R \times 25 \text{ J mol}^{-1}$
 (4) $2.303 \times R \times 298 \text{ J mol}^{-1}$
87. During compression of a spring the work done is 5 kJ and 1 kJ escaped to the surrounding as heat. The change in internal energy is:
 (1) 4 kJ
 (2) 6 kJ
 (3) -4 kJ
 (4) -6 kJ
88. Heat is absorbed when a gas 'x' (solute) is dissolved in water. The solubility of gas 'x' will increase at:
 (1) low temperature, low pressure
 (2) high temperature, High pressure
 (3) high temperature, low pressure
 (4) low temperature, High pressure
89. As per third law of thermodynamics:
 (1) $\lim_{T \rightarrow 0 \text{ K}} S \rightarrow 0$
 (2) $\lim_{T \rightarrow 0^\circ\text{C}} S \rightarrow 0$
 (3) $\lim_{T \rightarrow 0 \text{ K}} S \rightarrow \infty$
 (4) $\lim_{T \rightarrow 0^\circ\text{C}} S \rightarrow \infty$
90. The equilibrium constant for the reaction,
 $\text{A}(\text{s}) \rightleftharpoons \text{B}(\text{s}) + \frac{1}{2} \text{O}_2(\text{g})$ is $K_p = 9$. At equilibrium, the partial pressure of $\text{O}_2(\text{g})$ is:
 (1) 9 atm
 (2) 81 atm
 (3) 3 atm
 (4) 4.5 atm

91. In a cell the gene responsible for the expression of DNA ligase mutated leading to synthesis of faulty protein. What will be its direct consequence?
- (1) Synthesis of RNA from DNA will be hindered.
 - (2) An increase in the rate of transcription.
 - (3) The incorrect pairing of nitrogenous bases in the two strands of DNA.
 - (4) Accumulation of DNA fragments during replication.

92. Consider the following statements.
- A. Boulders grow by increase in mass.
 - B. Reproduction is seen in mules but not in sterile human couples.
 - C. Metabolism is seen only in eukaryotic cells, prokaryotic cells lack it.
 - D. Cellular organization is the defining feature of living organisms.
 - E. All non-living objects have the ability to sense the surroundings.
- In the light of above statements choose the **correct** answer from the options given below:
- (1) A, B and C are correct
 - (2) B, C and E are incorrect
 - (3) A, B, C and D are incorrect
 - (4) A, B, C, D and E are correct

93. In a polynucleotide chain of a DNA helix;
- (1) the backbone is made of pentose sugars, nitrogenous bases and phosphate groups.
 - (2) the backbone is made of phosphate groups bound to the nitrogenous bases.
 - (3) nitrogenous bases are present towards the inner side of the chain.
 - (4) phosphoester linkage is present between sugar and nitrogenous base.

94. The diagram below represents the regulation of the *lac* operon.



Choose the **correct** statement regarding it.

- (1) The regulation of *lac* operon by 'E' is referred to as negative regulation.
- (2) 'B, C and D' are required for metabolism of lactose.
- (3) 'A' is the product of one of the structural genes of the operon.
- (4) The operon consists of a monocistronic gene.

95. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Triticum</i>	(I)	Family
(B)	Anacardiaceae	(II)	Division
(C)	Poales	(III)	Genus
(D)	Angiospermae	(IV)	Order

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
 - (2) A-III, B-I, C-IV, D-II
 - (3) A-I, B-III, C-II, D-IV
 - (4) A-III, B-II, C-I, D-IV
96. Given below are two statements:
- Statement I:** In Meselson and Stahl's experiment, after one generation half the DNA molecules were heavy and half were hybrid or of intermediate density.
- Statement II:** The heavy isotope of nitrogen (^{15}N) was used to identify whether the genetic material is a protein or nucleic acid.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both Statement I and Statement II are correct.
 - (2) Both Statement I and Statement II are incorrect.
 - (3) Statement I is correct but Statement II is incorrect.
 - (4) Statement I is incorrect but Statement II is correct.

97. Identify the **incorrect** statement regarding the universal rules of binomial nomenclature.

- (1) The first word in a biological name represents the genus, and it starts with a capital letter.
- (2) Both the words in a biological name, when handwritten, are separately underlined.
- (3) The specific epithet starts with a small letter.
- (4) Biological names are generally in Greek and printed in bold to indicate their origin.

98. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The genetic code is nearly universal across all living organisms.

Reason (R): The genetic code is made up of three nitrogenous bases.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

99. An isolated metabolic reaction is performed in a test tube outside the body of an organism. This reaction is considered to be:

- (1) a living thing, as it is a metabolic reaction.
- (2) a non-living thing, but it is a living reaction.
- (3) neither living thing nor a living reaction, as it is outside a cell.
- (4) a non-living thing and a non-living reaction.

100. Arrange the following events that occur during the synthesis of a fully functional mRNA in a eukaryotic cell in the **correct** sequence.

- A. Removal of introns from primary transcript.
- B. Addition of a methyl guanosine triphosphate cap to primary transcript.
- C. Addition of adenylate residues at 3' end of hnRNA.
- D. Transport of functional mRNA out of the nucleus.
- E. Formation of heterogeneous nuclear RNA.

Choose the **correct** option.

- (1) E → B → C → A → D
- (2) A → B → C → E → D
- (3) B → E → A → C → D
- (4) C → B → A → D → E

101. Arrange the steps of DNA fingerprinting technique in the **correct** sequential order.

- A. Separation of DNA fragments by electrophoresis.
- B. Enzymatic digestion of DNA.
- C. Isolation of DNA.
- D. Hybridisation using labelled VNTR probe.

E. Transferring (blotting) of separated DNA fragments to synthetic membranes.

F. Detection of hybridised DNA fragments by autoradiography.

- (1) C → B → E → A → D → F
- (2) C → A → B → D → E → F
- (3) A → B → C → D → E → F
- (4) C → B → A → E → D → F

102. Which of the following pairs is correctly matched?

- (1) *Musca* and *Homo* – Present in different kingdoms
- (2) *Felis* and *Panthera* – Included in family Felidae
- (3) *Mangifera* and *Datura* – Members of family Solanaceae
- (4) *Solanum* and *sapiens* – Same level of taxa

103. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Hershey and Chase worked with organisms that infect bacteria.

Reason (R): The bacterial cell treats the viral genetic material as foreign and subsequently manufactures more virus particles.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

104. Choose the **correct** statement.

- (1) Genus is the lowest category of the taxonomical hierarchy.
- (2) The taxonomical hierarchy consists of 10 broad categories and 7 sub-categories.
- (3) As we go from base to the top of taxonomical hierarchy, the number of common characteristics amongst organisms does not change.
- (4) Division and phylum lie at the same level of taxonomical hierarchy.

- 105.** Identify the correct statements regarding the salient features of the Human Genome Project.
- The human genome contains 3164.7 million bp.
 - The average gene consists of 30,000 bases.
 - Less than 2% of the genome codes for proteins.
 - Chromosome Y has the most genes (2968).
 - Repetitive sequences make up a very small portion of the human genome.

Choose the **correct** option.

- A and C only
 - A, B and C
 - B, D and E
 - C and D only
- 106.** Given below are two statements:
- Statement I:** Two organisms present in same family will definitely be present in same genus.
- Statement II:** The basic requirement to place an organism in various categories is the knowledge of characters of an individual or group of organisms.
- In the light of the above statements, choose the **correct** answer from the options given below:
- Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
 - Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.

- 107.** Match **List-I** with **List-II** regarding a transcription unit.

List-I		List-II	
(A)	Promoter	(I)	Usually defines the end of the transcription process
(B)	Structural gene	(II)	DNA sequence that provides a binding site for RNA polymerase
(C)	Terminator	(III)	Does not code for anything but has a sequence same as RNA (except T at the place of U)
(D)	Coding strand	(IV)	The segment of DNA that is actually transcribed into RNA

Choose the *most appropriate* answer from the options given below.

- A-II, B-IV, C-I, D-III
- A-II, B-I, C-IV, D-III
- A-I, B-II, C-III, D-IV
- A-IV, B-II, C-I, D-III

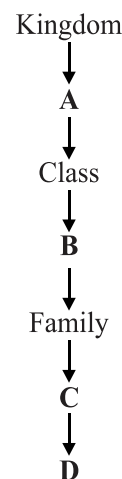
- 108.** Which of the following sets of species do not belong to the same kingdom?

- leo*, *pardus* and *tigris*
- nigrum*, *tuberosum* and *melongena*
- indica*, *aestivum* and *faba*
- elegans*, *coli* and *pneumoniae*

- 109.** Choose the **incorrect** statement about J. D. Watson.

- He was interested in bird-watching.
- He studied of the effect of hard X-rays on bacteriophage multiplication.
- Along with F. H. C. Crick, he was able to solve the DNA structure in the very first attempt.
- Watson and Crick proposed the complementary double-helical configuration of DNA in 1953.

- 110.** With respect to the taxonomical hierarchy shown below, choose the correct statement.



- More than one 'C' can be present in a 'D'.
- The animals classified in 'A' Arthropoda possess notochord and ventral hollow neural system.
- Monkey and Gorilla fall in category 'B' named Mammalia.
- Members of family Solanaceae and Convolvulaceae are grouped together in 'B' Polymoniales mainly based on floral characters.

- 111.** Identify the **correct** statement regarding a nucleosome.

- Eight acidic proteins constitute its core.
- The core is surrounded by a negatively charged molecule containing ribose sugar.
- It is the repeating unit of thread-like stained (coloured) bodies seen in nucleus.
- It is present only in prokaryotic cells.

112. Choose the **incorrect** statement.

- (1) The earliest classifications were based on the 'uses' of various organisms.
- (2) The word systematics is derived from the Latin word 'systema'.
- (3) Systematics takes into account evolutionary relationships between organisms.
- (4) 'Animals', 'mammals' and 'dogs' represent taxa at same levels.

113. Identify the **correct** statements regarding the process of translation.

- A. The ribosome in *E. coli* acts as a catalyst for peptide bond formation.
- B. UTRs are non-translating regions and hence have no role in translation.
- C. The process begins when the small subunit of the ribosome encounters an mRNA.
- D. In prokaryotes translation starts much before the transcription of the mRNA is completed.
- E. Aminoacylation of tRNA involves linking of activated amino acids to their cognate tRNA.

Choose the **correct** option.

- (1) A, B, C, E
- (2) B, C, D, E
- (3) A, B, C, D
- (4) A, C, D, E

114. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Properties of tissues are not present in the constituent cells but arise as a result of interactions among the constituent cells.

Reason (R): Properties of cellular organelles are not present in the molecular constituents of the organelle but arise as a result of interactions among the molecular components comprising the organelle.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

115. Given below is the sequence of a part of structural gene of a transcription unit:

3' AGTCAGTCAGTCAGTC 5'

If the underlined nitrogenous base is deleted, it will lead to;

- (1) a point mutation like sickle-cell anaemia.
- (2) no change in the number of codons.
- (3) change in the reading frame from that point.
- (4) no change in the number of amino acids being polymerised during translation.

116. Choose the **incorrect** statement about Carolus Linnaeus.

- (1) He proposed a system of providing a name with two components for living organisms.
- (2) He was the first person to describe *indica* species of mango.
- (3) He is called "The Darwin of the 20th century".
- (4) The title of his publication was *Systema Naturae*.

117. Given below are two statements:

Statement I: DNA is chemically less reactive and structurally more stable compared to RNA, making it a better genetic material.

Statement II: The 2'-OH group present at every nucleotide in RNA is a reactive group that makes RNA labile and easily degradable.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

118. Which of the following can multiply by fragmentation?

- A. Worker honey bees
- B. Filamentous algae
- C. Protonema of mosses
- D. Fungi
- E. *Hydra*

- (1) A and B only
- (2) B, C and D
- (3) C, D and E
- (4) A and C only

119. Match List-I with List-II.

List-I		List-II	
(A)	DNA fingerprinting	(I)	Based on DNA polymorphism
(B)	Expressed Sequence Tags	(II)	All the genes that are expressed as RNA
(C)	Restriction digestion	(III)	Cleavage of DNA by restriction endonucleases at specific sites
(D)	Variable number of tandem repeats	(IV)	Classified as mini-satellite DNA

Choose the *most appropriate* answer from the options given below.

- (1) A-II, B-I, C-IV, D-III
- (2) A-II, B-IV, C-I, D-III
- (3) A-I, B-II, C-III, D-IV
- (4) A-IV, B-II, C-I, D-III

120. In Griffith's experiment, which of the following lead to death of mice?

- (1) Injection of live R strain pneumococcus
- (2) Injection of heat-killed S strain with live R strain pneumococcus
- (3) Injection of heat-killed S strain pneumococcus
- (4) Injection of heat-killed R strain pneumococcus

121. All animals are placed in same;

- (1) Class.
- (2) Phylum
- (3) Kingdom.
- (4) Genus

122. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): For a double-stranded DNA, the ratios between Adenine and Thymine, and Guanine and Cytosine are constant and equal one.

Reason (R): In a DNA double helix, purines and pyrimidines are bonded to each other by peptide bonds.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

123. A member of family Canidae;

- (1) will be placed in class Insecta.
- (2) will not be placed in the kingdom in which wheat has been classified.
- (3) is placed in same order as *Petunia*.
- (4) is exactly similar to the organism with genus *Felis*.

124. Identify the **incorrectly matched pair.**

- (1) AUG - Initiator codon
- (2) UAA - Stop codon
- (3) GUG - Codes for Methionine
- (4) UGA - Stop codon

125. Match List-I with List-II.

List-I		List-II	
(A)	Genetic material in <i>E. coli</i>	(I)	3.3×10^9 bp
(B)	Genetic material in QB bacteriophage	(II)	48502 bp
(C)	Haploid content of human DNA	(III)	dsDNA
(D)	Bacteriophage lambda DNA	(IV)	RNA

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-I, C-III, D-IV

126. In the taxonomical hierarchy, the category Sapindales is placed between;

- (1) Monocotyledonae and Angiospermae
- (2) Dicotyledonae and Anacardiaceae
- (3) Monocotyledonae and Dicotyledonae
- (4) Dicotyledonae and Angiospermae

127. Read the following statements.

- A. RNA was the first genetic material.
- B. RNA contains 5-methyl uracil.
- C. RNA is one of the structural components of ribosome.
- D. RNA has evolved from DNA after chemical modifications.
- E. RNA is unable to mutate.

In the light of above statements choose the **correct** answer from the options given below:

- (1) A, C and E are correct
- (2) B, C and D are correct
- (3) B, D and E are incorrect
- (4) A, B and C are incorrect

128. Choose the **incorrect** statement.

- (1) Societies which indulged in anthropocentric view of biology could register limited progress in biological knowledge.
- (2) All present day living organisms are related to each other and also to all organisms that ever lived on this earth.
- (3) Early man could easily perceive the difference between inanimate matter and living organisms.
- (4) Early man deified some of the animate matter (wind, sea, fire etc.) and some among the animals and plants.

129. Given below are two statements:

Statement I: DNA-dependent DNA polymerases catalyse polymerisation only in the 5'→3' direction.

Statement II: On the template strand with polarity 5'→3', RNA synthesis is discontinuous.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

130. ICBN;

- (1) stands for Indian code for biological nomenclature.
- (2) has provided principles and criteria for scientific naming of plants.
- (3) is a group of animal taxonomists.
- (4) ensures that each organism has at least two names.

131. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Soluble RNA	(I)	mRNA
(B)	RNA that gets translated	(II)	23S rRNA
(C)	Ribozyme	(III)	18S rRNA
(D)	RNA transcribed by RNA polymerase I in nucleus	(IV)	tRNA

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-III, B-II, C-I, D-IV
- (4) A-IV, B-I, C-II, D-III

132. Read the following statements w.r.t DNA polymorphisms.

- A. It arises due to mutations.
- B. It is the basis of genetic mapping of the human genome.
- C. It is observed when an inheritable mutation occurs at a high frequency in a population.
- D. It is less likely to be observed in non-coding DNA sequences.
- E. It forms the basis of paternity testing.

The **true** statements are;

- (1) A, B, C, E
- (2) A, B, D only
- (3) B, C, D, E
- (4) A, D, E only

133. Why are both strands of a DNA segment not transcribed simultaneously?

- (1) It would lead to the formation of a double-stranded RNA, preventing translation.
- (2) The cell machinery is capable of reading only one strand at a time.
- (3) Transcription of both strands would produce identical proteins, which is inefficient.
- (4) The energy requirement for transcribing both strands is too high for the cell.

134. While studying the nucleus of cell, the scientist was able to count 50 beads on the 'beads-on-string' structure. Choose the correct statement about it.

- (1) The structure being referred to is the nucleolus.
- (2) The beads correspond to the nucleosomes.
- (3) As many as 1,00,000 bp are present in this structure.
- (4) The molecule being studied is a single stranded DNA.

135. According to the double-helix model of DNA;

- (1) the pitch of the helix is 0.34 nm.
- (2) the distance between a base pair in the helix is approximately 3.4 nm.
- (3) there are roughly 10 bp in each turn.
- (4) a purine on one strand is always bonded to a purine on the other strand.

136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: When the urinary bladder contracts to expel urine, the backflow of urine into the ureters is normally prevented.

Reason R: The act of urination involves the voluntary contraction of the sphincter muscles present in the urethra

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

137. Read the following statements regarding the principal nitrogenous wastes in animals.

- A. The common origin of ammonia, urea, and uric acid is the metabolic breakdown of nitrogen-containing macromolecules like proteins and nucleic acids.
- B. The amount of water required for the safe excretion of ammonia, urea, and uric acid is approximately the same for all three.
- C. The relative toxicity to the body is uniform across all three major forms of nitrogenous waste.
- D. In humans, the synthesis of urea occurs primarily in the cells of the kidney nephrons.

How many of the above statements is/are **incorrect**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

138. Given below are two statements.

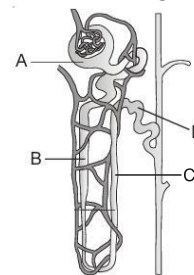
Statement I: Most nephrons in the kidney are cortical nephrons; they have short loops of Henle and are responsible for the bulk of urine formation under normal conditions.

Statement II: The high concentration of solutes in the kidney's medulla is partly maintained because urea is transported out of the collecting duct and re-enters the *vasa recta*.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

139. Which of the following labelled parts of nephron can be called as the diluting segment?



- (1) A
- (2) B
- (3) C
- (4) D

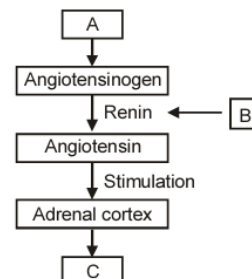
140. Match List-I with List-II.

List-I		List-II	
(A)	Mammals	(I)	Cockroaches
(B)	<i>Amphioxus</i>	(II)	Kidney
(C)	Malpighian tubules	(III)	Crustaceans
(D)	Antennal glands	(IV)	Protonephridia

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-IV, C-I, D-III

141. Identify the organs/chemicals labelled as A, B and C in the given flow chart of renin-angiotensin mechanism.



- (1) A - Kidney, B - Liver, C - Aldosterone
- (2) A - Kidney, B - Prorennin, C - Adrenaline
- (3) A - Liver, B - Prorennin C - ANF
- (4) A - Liver. B - Kidney. C - Aldosterone

142. Consider the following five statements (A-E) related to the excretory system.

- A. Filtration is a non-selective process performed by the glomerulus using the glomerular capillary blood pressure.
- B. Kidney has an inner convex surface and an outer concave surface.
- C. The collecting duct is not considered a part of the renal tubule of a nephron.
- D. Human kidneys can produce urine nearly four times more concentrated than the initial filtrate formed.
- E. In ammonotelic fishes, the kidneys do not play the major role in the removal of ammonia.

How many of the above statements are **correct**?

- (1) Two
- (2) Three
- (3) Four
- (4) Five

143. Which of the following statements **correctly** distinguishes the primary roles of the Loop of Henle and the *vasa recta* in the kidney's mechanism for concentrating urine?

- (1) The Loop of Henle acts as the exchanger by passively supplying blood to the medulla, while the *vasa recta* act as the multiplier by actively creating the osmotic gradient.
- (2) The Loop of Henle acts as the multiplier by using active transport to create the high solute concentration gradient in the medullary interstitium, while the *vasa recta* act as the exchanger to maintain this gradient by preventing the solutes from being washed away by blood flow.
- (3) Both the Loop of Henle and the *vasa recta* function as multipliers, working together to actively pump salt out of the filtrate and into the medullary interstitium.
- (4) The *vasa recta* act as the multiplier by establishing the initial gradient, while the Loop of Henle acts as the exchanger by fine-tuning the final concentration of urine through passive diffusion.

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Sweat gland	(I)	Most water-efficient excretory waste
(B)	Excess ingestion	(II)	Process of excreting ammonia
(C)	Ammonotelism	(III)	Eliminates urea, salts, and water through skin surface
(D)	Uric acid	(IV)	Leads to accumulation of ions like Na^+ , K^+ , Cl^- , phosphate, sulphate, etc., in animals

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-IV, C-II, D-I

145. Given below are two statements.

Statement I: The adult human kidney typically measures 10-12 cm in length and 5-7 cm in width, and is situated between the levels of the last thoracic and third lumbar vertebrae close to the ventral body wall, with the right kidney positioned slightly lower than the left.

Statement II: The process of defecation is not considered physiological excretion because it primarily involves the elimination of unabsorbed and undigested food residues that have remained within the alimentary canal and have not been produced by the body's metabolic activities.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

146. Arrange the following steps of haemodialysis in the **correct** logical sequence, from when blood is drawn to when it is returned to the body.

- A. To restore its normal clotting ability, the blood is mixed with an anti-heparin before being infused back into the patient via a vein.
- B. Blood is passed through a dialyzer, allowing urea and excess salts to diffuse out into a balanced dialysis fluid.
- C. To prevent clotting within the artificial kidney, an anticoagulant like heparin is added to the blood as it is drawn from an artery.

- (1) $C \rightarrow B \rightarrow A$ (2) $B \rightarrow C \rightarrow A$
- (3) $A \rightarrow B \rightarrow C$ (4) $C \rightarrow A \rightarrow B$

147. A patient with complete renal failure (end-stage renal disease) is undergoing regular haemodialysis. A blood test taken just before their dialysis session shows a very high concentration of blood urea. If this same patient were to suddenly suffer from acute liver failure, what would be the expected immediate trend in their blood urea concentration in the hours following the liver failure?

- (1) The blood urea concentration would continue to rise at an even faster rate.
- (2) The blood urea concentration would begin to fall rapidly as urea is metabolized by other tissues.
- (3) The blood urea concentration would stop increasing and plateau, remaining at its current high level.
- (4) The blood urea concentration would fluctuate unpredictably as both organs have failed.

148. Choose the option that **correctly** identifies the pair(s) as correct or incorrect.

(A)	CO_2 removed by lungs per minute	(I)	125 mL per minute
(B)	Glomerular filtration rate	(II)	200 mL per minute
(C)	Amount of urea excreted per day	(III)	70-80 gm
(D)	Number of nephrons per kidney	(IV)	Nearly one billion

- (1) Only pair A-I is correct
- (2) Only pair C-III is incorrect
- (3) Both B-II and D-IV are correct
- (4) All pairs are incorrect

149. A key distinction between the two types of nephrons in the human kidney is that juxtamedullary nephrons, are characterized by:
- (1) having their glomeruli located in the outer part of the renal cortex.
 - (2) being significantly smaller with a very short loop of Henle.
 - (3) a long loop of Henle extending deep into the medulla and an associated *vasa recta*.
 - (4) making up about 85% of the total nephrons in the kidney.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In a healthy individual, urine does not contain any glucose under normal conditions.

Reason R: Glucose is freely filtered from the blood into the glomerular filtrate, but it is 100% reabsorbed back into the blood via active transport in the proximal convoluted tubule (PCT).

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.
151. The micturition reflex is the process that controls urination. If, due to a specific neurological disorder, the stretch receptors in the wall of the urinary bladder become non-functional, what would be the most direct consequence?
- (1) The kidneys would receive a signal to stop producing urine.
 - (2) The muscle of the bladder would contract uncontrollably, leading to constant leakage.
 - (3) There would be no conscious sensation of bladder fullness, and the reflex for emptying the bladder would not be triggered.
 - (4) The brain would lose its voluntary control over the external urethral sphincter, making urination completely involuntary.

152. Consider the following statements regarding the functions and characteristics of Antidiuretic Hormone (ADH).

- A. When an individual drinks a large volume of water, the osmoreceptors in the hypothalamus are less stimulated, leading to a decrease in ADH release from the posterior pituitary.
- B. Exposure to cold temperatures can cause peripheral vasoconstriction, which increases central blood volume and suppresses ADH release, leading to diluted urine formation.

- C. During summer, heavy water loss by evaporation occurs in body, that suppresses releases of ADH
- D. ADH causes dilation of blood vessels, being a potent vasodilator.

How many of the above statements is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

153. Consider the following statements regarding the function of the Loop of Henle and its role in urine concentration.

- A. The descending limb is highly permeable to water but almost impermeable to electrolytes, causing the filtrate to become hypertonic as it descends into the medulla.
- B. The ascending limb is impermeable to water but allows for the transport of NaCl into the medullary interstitium, which is a key step in creating the osmotic gradient.
- C. As the filtrate passes up the ascending limb, it becomes progressively more concentrated because electrolytes are actively pumped into it from the medullary fluid.
- D. The ascending limb plays a minimal role in reabsorption but is critically important for maintaining the high osmolarity of the medullary interstitial fluid.

How many of the above statements is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

154. In the structure of the human kidney, the Columns of Bertini refer to the:

- (1) Conical sections of tissue that make up the renal medulla.
- (2) Extensions of the renal cortex that project down between the renal pyramids.
- (3) The tube that carries urine from the kidney to the bladder.
- (4) The outer fibrous layer that covers the kidney.

155. The Renin-Angiotensin system and the Atrial Natriuretic Factor (ANF) mechanism work antagonistically to regulate blood pressure. In a situation where a person's blood pressure is high, what physiological response would be expected?

- (1) The kidneys will secrete more renin, and the atria will stop releasing ANF.
- (2) The atria of the heart will release ANF, which causes vasodilation to lower blood pressure.
- (3) The atria of the heart will release ANF, which causes vasoconstriction to further increase blood pressure.
- (4) Both renin and ANF will be released simultaneously to stabilize the blood pressure.

156. Consider the following statements regarding the accessory excretory organs in the human body.

- A. The liver contributes to excretion by secreting substances like bilirubin, degraded steroid hormones, and cholesterol into bile, which are then eliminated with digestive wastes.
- B. The secondary role of sweat glands is providing a cooling effect on the body surface.
- C. Sebaceous glands secrete sebum, which contains sterols and waxes, providing a protective oily layer for the skin and aiding in the elimination of these substances.
- D. In addition to other organs, small quantities of nitrogenous wastes can also be eliminated from the body through saliva.

How many of the above statements is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

157. Select the **correct** pair representing an organism and its primary nitrogenous waste product.

- (1) Marine cartilaginous fish → Ammonia
- (2) Terrestrial amphibians → Urea
- (3) Bony fishes → Urea
- (4) Reptiles → Urea

158. Given below are two statements.

Statement I: The peritubular capillary network is associated with the renal tubules of all nephrons, while the *vasa recta* is a specialized capillary network found exclusively in association with the a specific type of nephron.

Statement II: The renal calyces are cup-like structures that collect urine draining from the renal pyramids, which in turn merge to form the larger, funnel-shaped renal pelvis that funnels urine into the ureter.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

159. The kidney's primary function is to filter metabolic wastes from the blood. Considering the blood vessels that enter and exit the kidney at the hilum, which one would have the minimum concentration of urea?

- (1) Renal Artery
- (2) Renal Vein
- (3) Abdominal Aorta
- (4) Inferior Vena Cava

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The presence of significant levels of ketone bodies in the urine, a condition known as ketonuria, is a characteristic finding in individuals with uncontrolled diabetes mellitus.

Reason R: In diabetes mellitus the glucose present in blood is not utilised which forces the body to switch to the rapid and excessive metabolism of fats, leading to the overproduction of ketone bodies by the liver.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

161. The Glomerular Filtration Rate (GFR) in a healthy adult is more than 150 liters per day. However, the average daily urine output is only about 1.5 liters. What is the primary reason for this significant difference in volume?

- (1) Most of the filtered water is lost as sweat or through respiration.
- (2) The urinary bladder is capable of reabsorbing the vast majority of the water before urination.
- (3) Approximately 99% of the water and useful solutes from the glomerular filtrate are reabsorbed by the renal tubules.
- (4) The process of tubular secretion removes water from the filtrate and returns it to the blood.

162. Given below are two statements.

Statement I: Aldosterone stimulates active Na⁺ reabsorption in the distal tubule and collecting duct; water reabsorption follows passively, provided an osmotic gradient exists.

Statement II: Osmoregulation is the active physiological process by which an organism controls the water balance and solute (like salt) concentrations in its body to maintain a stable internal environment.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

163. A healthy individual rapidly consumes one liter of 0.9% saline solution, which is isotonic to their blood plasma. What would be the expected changes in their urine volume and the concentration of Na^+ ions in their urine, compared to their normal physiological state?

- (1) Urine volume will not change, and the concentration of Na^+ will decrease (dilute urine).
- (2) Urine volume will increase, and the concentration of Na^+ will also increase.
- (3) Urine volume will decrease, and the concentration of Na^+ will increase (concentrated urine).
- (4) Both urine volume and the concentration of Na^+ will remain unchanged.

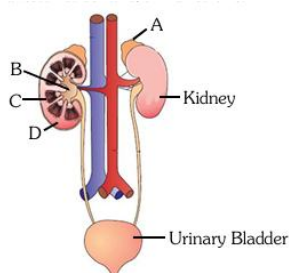
164. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Urine	(I)	Has a characteristic odour
(B)	Brush bordered epithelium	(II)	PCT
(C)	Glomerulus	(III)	A tuft of capillaries
(D)	Afferent arteriole	(IV)	Fine branch of renal artery

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-IV, B-I, C-III, D-II
- (3) A-I, B-II, C-III, D-IV
- (4) A-I, B-II, C-IV, D-III

165. Examine the provided diagram of the human excretory system. Which of the following options provides the **correct** identification of a labelled part along with its accurate description?



- (1) **A** - Adrenal gland; a suprarenal gland that secretes aldosterone
- (2) **B** - Renal pelvis; the site of ultrafiltration where the primary urine is formed from blood.
- (3) **C** - Medulla; the region containing the Malpighian corpuscles and responsible for selective reabsorption of glucose and amino acids.
- (4) **D** - Cortex; the innermost part of the kidney, characterized by its striated appearance due to the presence of collecting ducts.

166. The juxtaglomerular (JG) apparatus is a specialized regulatory structure in the nephron. It is formed by the close association of which two components?

- (1) The macula densa of the distal convoluted tubule and the juxtaglomerular cells of the afferent arteriole.
- (2) The podocytes of the Bowman's capsule and the cells of the proximal convoluted tubule.
- (3) The macula densa of the distal convoluted tubule and the smooth muscle cells of the efferent arteriole.
- (4) The juxtaglomerular cells of the afferent arteriole and the epithelial cells of the collecting duct.

167. Consider the following physiological effects attributed to the hormone Angiotensin II.

- A. It constricts the efferent arteriole, which can help maintain the Glomerular Filtration Rate (GFR).
- B. It is a potent vasoconstrictor, which leads to an increase in systemic blood pressure.
- C. It enhances the reabsorption of Na^+ and water from the distal convoluted tubule by stimulating secretion of aldosterone.
- D. It causes a decrease in blood volume by promoting the excretion of water and Na^+
- E. It is secreted by JG cells.

How many of the above are **correct**.

- (1) Three
- (2) Four
- (3) Five
- (4) Two

168. Which of the following statements provides the most **accurate** description of where the key processes of urine formation occur within a nephron?

- (1) Glomerular filtration is confined to the renal corpuscle, while tubular reabsorption and secretion happen at various points along the renal tubule.
- (2) The majority of both filtration and reabsorption is completed within the Proximal Convoluted Tubule (PCT).
- (3) The glomerulus is responsible for filtration, the Loop of Henle is responsible for reabsorption, and the Distal Convoluted Tubule (DCT) is responsible for secretion.
- (4) Tubular secretion moves substances from the filtrate into the blood, while reabsorption moves them from the blood into the filtrate.

169. The volume of blood that flows through the kidneys per minute is **best** represented as:

- (1) a significant fraction (~20-25%) of the total cardiac output.
- (2) a volume nearly equal to the entire cardiac output.
- (3) the same volume as the glomerular filtrate produced per minute.
- (4) an amount far greater than the total cardiac output.

170. Select the **incorrect** match.

- (1) DCT – Facultative water absorption
- (2) Ureter– Carries urine from kidney to urinary bladder
- (3) Transitional epithelium – Present in whole excretory system
- (4) Bowman's capsule – Double walled cup-like structure

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Zones of kidney	(I)	Reabsorbed in PCT and DCT
(B)	Loop of Henle	(II)	Inner medulla and outer cortex
(C)	<i>Vasa recta</i>	(III)	Hairpin shaped
(D)	HCO_3^-	(IV)	'U' shaped

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV D-III
- (2) A-II, B-III, C-IV, D-I
- (3) A-II, B-I, C-III, D-IV
- (4) A-II, B-IV, C-I, D-III

172. Read the following statements (A to D) and choose the option which states them as true (T) or False (F) **correctly**.

- A. Nearly all of the essential nutrients, such as glucose and amino acids, are reabsorbed by the Proximal Convolute Tubule (PCT).
- B. Uric acid is a metabolic waste product that is highly soluble in water.
- C. Renal pyramids are primarily composed of the parallel arrangement of the loops of Henle and the collecting ducts.
- D. The Proximal and Distal convolute tubules are located in the renal cortex and are not a part of the renal pyramids.

A B C D

- (1) T T F F
- (2) T F T T
- (3) F F T F
- (4) T T T F

173. Which of the following statements **most accurately** describes the quantitative relationship between the functional units of the kidney and their primary components?

- (1) The total number of nephrons is equivalent to the total number of renal corpuscles (Malpighian bodies).
- (2) The number of nephrons is half the total number of glomeruli, as each glomerulus serves two nephrons.
- (3) The number of nephrons is more than the sum of all Bowman's capsules and all associated renal tubules.
- (4) The number of nephrons is equal to the number of collecting ducts in a kidney.

174. For blood to be filtered in the kidney, plasma fluid must pass through a specialized three-layered structure known as the filtration membrane. Which of the following **correctly** lists these three layers in the order that filtrate passes through them?

- (1) The fenestrated (having pores) endothelium of the glomerular capillary, the basement membrane, and the filtration slits of the podocytes.
- (2) The filtration slits of the podocytes, the basement membrane, and the fenestrated (having pores) endothelium of the glomerular capillary.
- (3) The parietal layer of the Bowman's capsule, the basement membrane, and the endothelium of the glomerular capillary.
- (4) The basement membrane, the macula densa cells, and the fenestrated endothelium of the glomerular capillary.

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Platyhelminthes	(I)	Nephridia
(B)	Vertebrates	(II)	Flame cells
(C)	Earthworm	(III)	Complex tubular forms of excretory structures
(D)	Renal capsule	(IV)	Tough protective covering around kidney

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-IV, C-I, D-III

176. Which of the following statements **best** describes the condition of Diabetes Insipidus and its underlying cause?

- (1) The excretion of large volumes of urine containing glucose due to the hyposecretion of insulin.
- (2) The excretion of large volumes of dilute urine due to the hyposecretion of Antidiuretic Hormone (ADH).
- (3) The excretion of a small volume of concentrated urine due to the hypersecretion of Antidiuretic Hormone (ADH).
- (4) The excretion of large volumes of urine with high sodium content due to the hyposecretion of aldosterone.

177. Consider the following statements related to the human excretory system.

- A. Renin is a hormone.
- B. The characteristic yellow color of normal urine is due to the presence of a pigment called urochrome.
- C. Diuresis is the medical term for the condition of frequent urination.
- D. A reduced intake of protein-rich food can lead to a lower total amount of urea being excreted in the urine.
- E. Renal calculi are insoluble masses of crystallized salts, commonly known as kidney stones, that can form within the kidney.

Which combination of the above statements is **correct**?

- (1) A, C and E only
- (2) A and C only
- (3) B, D and E only
- (4) B and D only

178. Given below are two statements.

Statement I: The Malpighian corpuscle consists of the glomerulus surrounded by the Bowman's capsule.

Statement II: The surgical removal of one kidney is fatal due to an immediate failure to manage the body's waste products.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

■■■

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: A high hydrostatic pressure is generated within the glomerular capillaries, which provides the primary driving force for the process of ultrafiltration.

Reason R: The efferent arteriole, which carries blood away from the glomerulus, has a significantly broader diameter than the afferent arteriole, which brings blood to the glomerulus.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

180. The pH of normal human urine is typically slightly acidic, ranging from 4.5 to 8.0 but averaging around 6.0. What is the primary physiological mechanism responsible for this acidity?

- (1) The filtration of acidic proteins from the blood into the Bowman's capsule.
- (2) The reabsorption of water in the collecting duct, which concentrates the filtrate.
- (3) The tubular secretion of hydrogen ions (H^+) into the filtrate and reabsorption of bicarbonate ions (HCO_3^-).
- (4) The breakdown of urea into ammonia within the renal tubule, which lowers the pH.

SPACE FOR ROUGH WORK



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>

Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____